# PETRA ZLATARIĆ-NOVOSEL

## PROFESSIONAL SUMMARY

Machine Learning Engineer with experience developing and operating NLP and document AI systems for Croatian-language public administration documents. Delivered document classification serving approximately 15,000 documents per day, built a Croatian named entity recognition model, and implemented model drift monitoring that helped identify and resolve an OCR-related accuracy drop. Hands-on experience with Python, PyTorch, Hugging Face Transformers, FastAPI, Docker, Azure and MLflow. Also built a personal, non-production retrieval-augmented question-answering demo over Croatian laws.

## TECHNICAL SKILLS

**Programming and data:** Python, pandas, Oracle SQL, PostgreSQL
**Machine learning and NLP:** PyTorch, scikit-learn, Hugging Face Transformers, spaCy, fastText, sentence-transformers, text classification, named entity recognition, TF-IDF, retrieval-augmented generation (RAG)
**Deployment and MLOps:** FastAPI, Docker, Azure App Service, Azure ML, Azure Functions, MLflow, Evidently, model serving, model drift monitoring
**Data annotation:** Label Studio, annotation guidelines, inter-annotator agreement

## PROFESSIONAL EXPERIENCE

**ML Engineer | Slavonica Document AI** *2023–Present*

- Develop document classification and named entity recognition solutions for Croatian public administration documents.
- Built and deployed a scanned-document classifier for 23 classes, achieving 94% accuracy and a macro F1 score of 0.91.
- Fine-tuned XLM-R base and served the classifier as a FastAPI service in Docker on Azure App Service, processing approximately 15,000 documents per day.
- Developed a Croatian NER model to identify persons, organisations, OIB numbers, addresses and legal references in administrative text.
- Created an annotated dataset of 4,000 documents, including an annotation guideline; coordination across two annotators achieved an inter-annotator kappa of 0.87.
- Implemented weekly drift reporting for text length, vocabulary and predicted-class distribution, with alerts when PSI exceeded 0.2. Identified a silent client OCR change associated with a 9-point accuracy drop; the issue was fixed within a week.

**Data Scientist | Baranjska kreditna banka** *2021–2023*

- Developed churn prediction models and complaint classification solutions.
- Built a customer complaint topic classifier, progressing from a TF-IDF and logistic regression baseline to fastText across 11 topics.
- Delivered weekly topic reporting for customer service management.

## SELECTED PROJECT

**Retrieval-Augmented Question-Answering Demo | Personal project** *2025*

- Built a weekend prototype for question answering over Croatian laws using a sentence embedding model and an open-weight LLM accessed through an API.

- Implemented retrieval with FAISS; the prototype was not used in production and was checked manually without formal evaluation.

## EDUCATION

Doctoral studies, named entity recognition in Croatian legal texts | University of Drava | Expected 2027

## TEACHING AND PUBLICATION

Guest lecturer, NLP Summer School | 2025 | 90-minute lecture
Croatian Named Entity Recognition research published at a regional workshop | 2024

## LANGUAGES

English: Used daily in meetings with a client in Slovenia